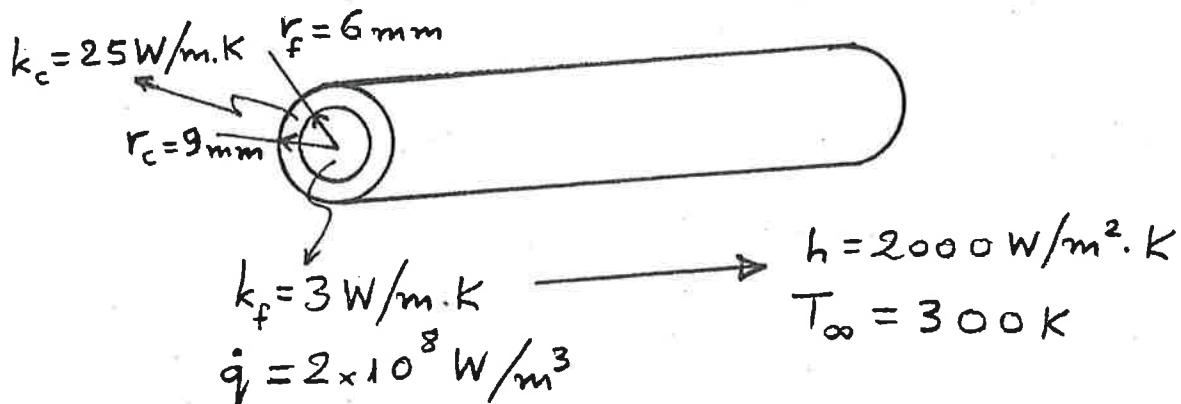


COURSE: CHE 312: Transport Phenomena II**Spring Semester 2020****Department of Chemical Engineering****University of Illinois at Chicago****Chicago, Illinois 60607-7052****Quiz #2****February 12, 2020*****Closed Notes and Book*****Duration: 25 minutes**

6-5/10

A nuclear reactor fuel element consists of a solid cylindrical pin of radius r_f and thermal conductivity k_f . The fuel pin is in good contact with a cladding material of outer radius r_c and thermal conductivity k_c . Consider steady-state conditions for which uniform heat generation occurs within the fuel at a volumetric rate $\dot{q}$ and the outer surface of the cladding is exposed to a coolant that is characterized by a temperature T_∞ and a convection coefficient h .

- Obtain equations for the temperature distributions $T_f(r)$ and $T_c(r)$ in the fuel and cladding, respectively. Express your results exclusively in terms of the foregoing variables.
- Consider a uranium oxide fuel pin for which $k_f = 3 \text{ W/m}\cdot\text{K}$ and $r_f = 6 \text{ mm}$ and cladding for which $k_c = 25 \text{ W/m}\cdot\text{K}$ and $r_c = 9 \text{ mm}$. If $\dot{q} = 2 \times 10^8 \text{ W/m}^3$, $h = 2000 \text{ W/m}^2\cdot\text{K}$, and $T_\infty = 300 \text{ K}$, what is the maximum temperature in the fuel element?



for $0 < r < r_c$

Assumptions

$$\frac{\partial^2 T_c}{\partial r^2} + \frac{\dot{q}}{k} = 0$$

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T_c}{\partial r} \right) = -\frac{\dot{q}}{k_c}$$

$$\frac{\partial^2 T_c}{\partial r^2} = -\frac{\dot{q}}{k}$$

$$\frac{\partial}{\partial r} \left(r \frac{\partial T_c}{\partial r} \right) = -\frac{\dot{q}}{k_c} r$$

$$\frac{\partial T_c}{\partial r} =$$

$$r \frac{dT_c}{dr} = -\frac{\dot{q}}{2k_c} r^2 + C_1$$

$$\frac{dT_c}{dr} = -\frac{\dot{q}}{2k_c} r + \frac{C_1}{r}$$

$$T_c(r) = -\frac{\dot{q}}{4k_c} r^2 + C_1 \ln(r) + C_2$$

@ $T_c(0)$, T is finite

$$T_c(0) = 0 + C_1 \ln(0) + C_2$$

$$T_c(0) = C_1 + C_2$$

$$@ T_c(r_c) = T_f(r_c)$$

$r_c < r < r_f$

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T_f}{\partial r} \right) = 0$$

$$r \frac{dT_f}{dr} = C_3$$

$$\frac{dT_f}{dr} = \frac{C_3}{r}$$

$$T_f(r) = C_3 \ln(r) + C_4$$

$$-k_f \frac{\partial T_f}{\partial r} \bigg|_{r=r_f} = h [T_{\infty} - T_f]$$

$$\frac{dT_f}{dr} \bigg|_{r=r_f} = \frac{C_3}{r} \bigg|_{r=r_f}$$

$$-k_f \frac{dT_f}{dr} \bigg|_{r=r_f} = -k_f \frac{C_3}{r_f}$$

$$h [T_{\infty} - T_f] = -k_f \frac{C_3}{r_f}$$

$$-\frac{h r_f}{k_f} [T_{\infty} - T_f] = C_3$$

$$q_f'' = -k_f \frac{\partial T_f}{\partial r} = (-k_f) \frac{h r_f}{k_f} [T_{\infty} - T_f] \left(\frac{1}{r} \right)$$

$$q_f'' = \frac{h r_f}{r} (T_{\infty} - T_f)$$

$$q_c'' = -k_c \frac{dT_c}{dr} = -k_c \left(-\frac{\dot{q}}{2k_c} r + \frac{T_c(0) - C_2}{r} \right)$$

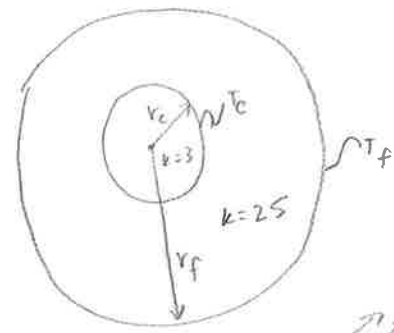
$$q_c'' = \frac{\dot{q} r}{2} - \frac{k_c (T_c(0) - C_2)}{r}$$

$$\frac{h r_f}{r} (T_{\infty} - T_f) = \frac{\dot{q} r}{2} - \frac{C_1 k_c}{r}$$

$$h r_f (T_{\infty} - T_f) = \frac{\dot{q} r^2}{2} - C_1 k_c$$

$$C_1 = \frac{h r_f (T_{\infty} - T_f) - \frac{\dot{q} r^2}{2}}{k_c}$$

Look at previous
page for more
progress



T_{∞}
 h

$$T_c(r) = -\frac{\dot{q}}{4k_c} r^2 + C_1 \ln(r) + C_2$$

$$T_f(r) = C_3 \ln(r) + C_4$$

$$@ T_c(r_c) = T_f(r_c) \text{ solve for } C_4$$

$$-\frac{\dot{q}}{4k_c} (r_c)^2 - \frac{h r_f (T_{\infty} - T_f) - \frac{\dot{q} r^2}{2}}{k_c} (\ln(r_c) + 1) - T_c(0)$$

$$= -\frac{h r_f}{k_f} (T_{\infty} - T_f) \ln(r_c) + C_4$$